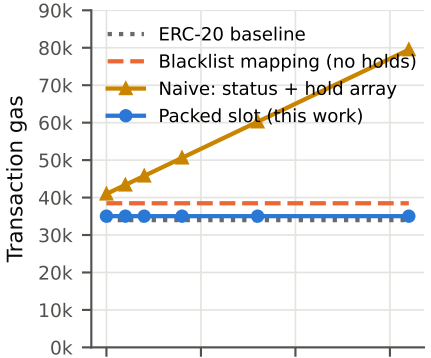
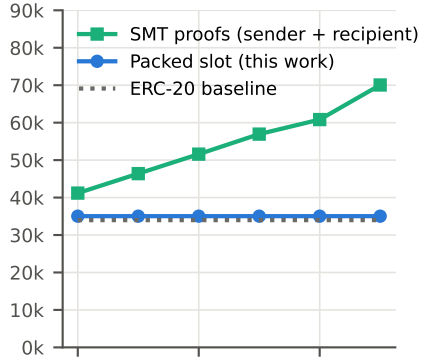
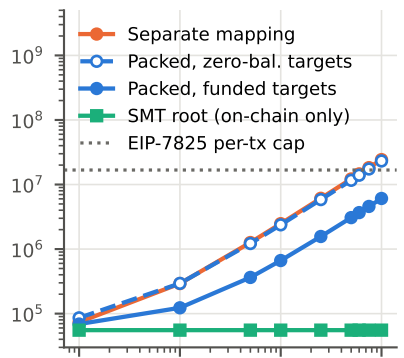

(a) Transfer cost vs. holds
ERC-20 baseline
Blacklist mapping (no holds)
Naive: status + hold array
Packed slot (this work)
Transaction gas
90k
80k
70k
60k
50k
40k
30k
20k
10k
0k
0
5
10
15
Active holds on sender k


(b) SMT transfer cost vs. list size
SMT proofs (sender + recipient)
Packed slot (this work)
ERC-20 baseline
90k
80k
70k
60k
50k
40k
30k
20k
10k
0k
10^0
10^2
10^4
Restricted addresses N


(c) List update cost (2-of-3)
Separate mapping
Packed, zero-bal. targets
Packed, funded targets
SMT root (on-chain only)
EIP-7825 per-tx cap
10^9
10^8
10^7
10^6
10^5
10^0
10^1
10^2
10^3
Batch size K